

Table 1: TruthfulQA: success (%). Bold denotes all tied numerical maxima. Counts and provenance accompany the generated CSV.

	Direct	SR	Rfx	FR	ReAct	SChk	ToT	INTRO-SPECTER
Llama 8B	80.6	61.1	94.4	86.1	72.2	91.7	77.8	94.4
Llama 70B	83.3	61.1	91.7	80.6	75.0	83.3	80.6	94.4
Mistral 12B	63.9	61.1	80.6	69.4	63.9	72.7	56.5	86.1

Table 2: 2WikiMultiHopQA: success (%). Bold denotes all tied numerical maxima. Counts and provenance accompany the generated CSV.

	Direct	SR	Rfx	FR	ReAct	SChk	ToT	INTRO-SPECTER
Llama 8B	86.7	76.7	93.3	78.3	76.7	66.7	80.0	96.7
Llama 70B	96.7	91.7	96.7	95.0	95.0	93.3	90.0	96.7
Mistral 12B	91.7	86.7	95.0	90.0	86.7	89.8	92.2	100.0

Jazz results review: scope 12

Across 12 cells, INTRO-SPECTER records 579/648 successes (89.35%). On 648 matched observations, the IS-Reflexion difference is +2.62 pp (89.35% versus 86.73%). The cell record against Reflexion is 8 numerical wins, 2 ties, and 2 losses; 0 wins and 0 losses survive Holm adjustment. Only the three runnable models have SPR reruns. Qwen archival no-SPR results are excluded from this scope. Baselines are archival; paired task/seed keys do not remove provider drift. Missing observations are not zero scores.

Table 3: LongMemEval: success (%). Bold denotes all tied numerical maxima. Counts and provenance accompany the generated CSV.

	Direct	SR	Rfx	FR	ReAct	SChk	ToT	INTRO-SPECTER
Llama 8B	63.3	51.7	76.7	63.3	61.7	50.0	55.0	88.3
Llama 70B	43.3	48.3	70.0	45.0	40.0	33.3	30.0	60.0
Mistral 12B	55.0	50.0	73.3	58.3	58.3	53.3	67.9	81.7

Table 4: MuSiQue: success (%). Bold denotes all tied numerical maxima. Counts and provenance accompany the generated CSV.

	Direct	SR	Rfx	FR	ReAct	SChk	ToT	INTRO-SPECTER
Llama 8B	76.7	66.7	90.0	76.7	66.7	51.7	71.7	95.0
Llama 70B	83.3	78.3	96.7	81.7	81.7	70.0	55.0	93.3
Mistral 12B	71.7	61.7	85.0	71.7	68.3	75.0	68.6	88.3

Table 5: Cell outcomes against Reflexion; wins/ties are numerical, loss significance uses Holm adjustment.

Dataset	Wins	Ties	NS losses	Sig. losses
TruthfulQA	2	1	0	0
2WikiMultiHopQA	2	1	0	0
LongMemEval	2	0	1	0
MuSiQue	2	0	1	0

Table 6: TruthfulQA: exact raw p , paired-bootstrap 95% CI (pp), and paired n in parentheses. *: Holm-adjusted $p < .05$ within the baseline family; sign of the CI determines direction.

	Direct	SR	Rfx	FR	ReAct
Llama 8B	0.062 [+2.8,+25.0] (36)	< .001* [+19.4,+50.0] (36)	1.000 [+0.0,+0.0] (36)	0.250 [+0.0,+19.4] (36)	0.008* [+8.3,+36.1] (36)
Llama 70B	0.125 [+2.8,+22.2] (36)	< .001* [+19.4,+50.0] (36)	1.000 [+0.0,+8.3] (36)	0.062 [+2.8,+25.0] (36)	0.016* [+8.3,+33.3] (36)
Mistral 12B	0.021 [+5.6,+38.9] (36)	0.022 [+8.3,+41.7] (36)	0.625 [-5.6,+16.7] (36)	0.031 [+5.6,+30.6] (36)	0.008* [+8.3,+36.1] (36)

Table 7: 2WikiMultiHopQA: exact raw p , paired-bootstrap 95% CI (pp), and paired n in parentheses. *: Holm-adjusted $p < .05$ within the baseline family; sign of the CI determines direction.

	Direct	SR	Rfx	FR	ReAct
Llama 8B	0.031 [+3.3,+18.3] (60)	< .001* [+10.0,+30.0] (60)	0.625 [-3.3,+10.0] (60)	< .001* [+8.3,+28.3] (60)	< .001* [+10.0,+30.0] (60)
Llama 70B	1.000 [+0.0,+0.0] (60)	0.250 [+0.0,+11.7] (60)	1.000 [+0.0,+0.0] (60)	1.000 [-3.3,+8.3] (60)	1.000 [+0.0,+5.0] (60)
Mistral 12B	0.062 [+1.7,+16.7] (60)	0.008* [+5.0,+23.3] (60)	0.250 [+0.0,+11.7] (60)	0.031 [+3.3,+18.3] (60)	0.008* [+5.0,+21.7] (60)

Table 8: LongMemEval: exact raw p , paired-bootstrap 95% CI (pp), and paired n in parentheses. *: Holm-adjusted $p < .05$ within the baseline family; sign of the CI determines direction.

	Direct	SR	Rfx	FR	ReAct
Llama 8B	< .001* [+13.3,+36.7] (60)	< .001* [+25.0,+48.3] (60)	0.065 [+1.7,+21.7] (60)	< .001* [+13.3,+36.7] (60)	< .001* [+15.0,+30.0] (60)
Llama 70B	0.006 [+6.7,+28.3] (60)	0.092 [+0.0,+23.3] (60)	0.210 [-23.3,+1.7] (60)	0.004* [+6.7,+25.0] (60)	0.002* [+10.0,+30.0] (60)
Mistral 12B	< .001* [+16.7,+38.3] (60)	< .001* [+20.0,+43.3] (60)	0.227 [-1.7,+18.3] (60)	< .001* [+13.3,+35.0] (60)	< .001* [+11.7,+30.0] (60)

Table 9: MuSiQue: exact raw p , paired-bootstrap 95% CI (pp), and paired n in parentheses. *: Holm-adjusted $p < .05$ within the baseline family; sign of the CI determines direction.

	Direct	SR	Rfx	FR	ReAct
Llama 8B	< .001* [+10.0,+28.3] (60)	< .001* [+16.7,+40.0] (60)	0.375 [-1.7,+11.7] (60)	0.003* [+8.3,+30.0] (60)	< .001* [+16.7,+40.0] (60)
Llama 70B	0.070 [+1.7,+20.0] (60)	0.012* [+5.0,+25.0] (60)	0.625 [-10.0,+3.3] (60)	0.039 [+3.3,+21.7] (60)	0.039 [+3.3,+21.7] (60)
Mistral 12B	0.013 [+5.0,+28.3] (60)	< .001* [+13.3,+40.0] (60)	0.754 [-6.7,+13.3] (60)	0.021 [+5.0,+28.3] (60)	0.002* [+10.0,+30.0] (60)

Table 10: Recorded mean tokens per observed task; ratios use the recorded IS mean as denominator. Token accounting is inherited from the released harness; archival baselines and SPR reruns have different provider snapshots.

Dataset	Direct	ratio	Rfx	ratio	SChk	ratio	ToT	ratio	IS	ratio
TruthfulQA	1202	0.39	2380	0.78	5901	1.94	34270	11.25	3047	1.00
2WikiMultiHopQA	2034	0.42	2767	0.57	10341	2.12	58284	11.95	4878	1.00
LongMemEval	2204	0.44	6302	1.25	12179	2.41	69104	13.70	5043	1.00
MuSiQue	2142	0.42	4046	0.79	11254	2.18	71915	13.96	5152	1.00